



PDF Download
3576915.3624396.pdf
25 December 2025
Total Citations: 2
Total Downloads: 273

Latest updates: <https://dl.acm.org/doi/10.1145/3576915.3624396>

POSTER

Poster: Detecting Adversarial Examples Hidden under Watermark Perturbation via Usable Information Theory

ZIMING ZHAO, Zhejiang University, Hangzhou, Zhejiang, China

ZHAOXUAN LI, Institute of Information Engineering, Beijing, China

TINGTING LI, Zhejiang University, Hangzhou, Zhejiang, China

ZHUOXUE SONG, Zhejiang University, Hangzhou, Zhejiang, China

FAN ZHANG, Zhejiang University, Hangzhou, Zhejiang, China

RUI ZHANG, Institute of Information Engineering, Beijing, China

Open Access Support provided by:

Zhejiang University

Institute of Information Engineering

Published: 15 November 2023

[Citation in BibTeX format](#)

CCS '23: ACM SIGSAC Conference on
Computer and Communications Security
November 26 - 30, 2023
Copenhagen, Denmark

Conference Sponsors:
SIGSAC

Poster: Detecting Adversarial Examples Hidden under Watermark Perturbation via Usable Information Theory

Ziming Zhao
Zhejiang University
Hangzhou, China
zhaoziming@zju.edu.cn

Zhaoxuan Li
Institute of Information Engineering,
Chinese Academy of Sciences, China
lizhaoxuan@iie.ac.cn

Tingting Li
Zhejiang University
Hangzhou, China
litt2020@zju.edu.cn

Zhuoxue Song
Zhejiang University
Hangzhou, China
songzhuoxue@zju.edu.cn

Fan Zhang*
Zhejiang University
Hangzhou, China
fanzhang@zju.edu.cn

Rui Zhang
Institute of Information Engineering,
Chinese Academy of Sciences, China
zhangrui@iie.ac.cn

ABSTRACT

Image watermark is a technique widely used for copyright protection. Recent studies show that the image watermark can be added to the clear image as a kind of noise to realize fooling deep learning models. However, previous adversarial example (AE) detection schemes tend to be ineffective since the watermark logo differs from typical noise perturbations. In this poster, we propose Themis, a novel AE detection method against watermark perturbation. Different from prior methods, Themis neither modifies the protected classifier nor requires knowledge of the process for generating AEs. Specifically, Themis leverages usable information theory to calculate the pointwise score, thereby discovering those instances that may be watermark AEs. The empirical evaluations involving 5 different logo watermark perturbations demonstrate the proposed scheme can efficiently detect AEs, and significantly (over 15% accuracy) outperforms five state-of-the-art (SOTA) detection methods. The visualization results display our detection metric is more distinguishable between AEs and non-AEs. Meanwhile, Themis realizes a larger Area Under Curve (AUC) in a threshold-resilient manner, while only introducing $\sim 0.04s$ overhead.

CCS CONCEPTS

• Security and privacy $\rightarrow$ Domain-specific security and privacy architectures.

KEYWORDS

Watermark Adversarial Examples, Usable Information Theory

ACM Reference Format:

Ziming Zhao, Zhaoxuan Li, Tingting Li, Zhuoxue Song, Fan Zhang, and Rui Zhang. 2023. Poster: Detecting Adversarial Examples Hidden under Watermark Perturbation via Usable Information Theory. In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*

*Corresponding author.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CCS '23, November 26–30, 2023, Copenhagen, Denmark

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0050-7/23/11.

<https://doi.org/10.1145/3576915.3624396>

(CCS '23), November 26–30, 2023, Copenhagen, Denmark. ACM, Copenhagen, Denmark, 3 pages. <https://doi.org/10.1145/3576915.3624396>

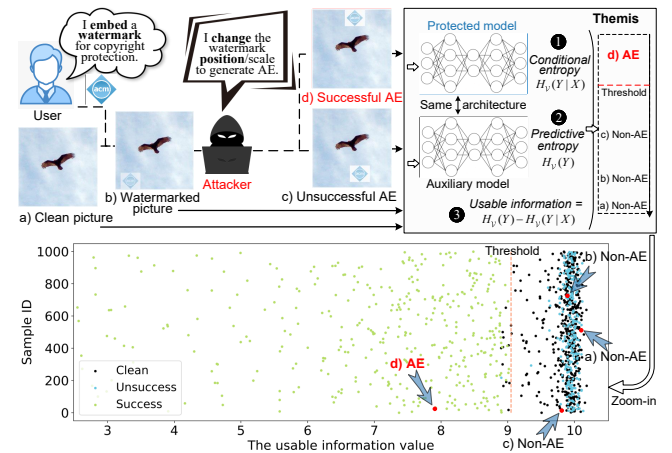


Figure 1: Illustrative explanation of Themis. The usable information value calculated by Themis can be used to distinguish AE & non-AE.

1 INTRODUCTION

Watermarking technology plays an important role in protecting intellectual property rights. However, recent research, Adv-watermark [4], reveals that the visible watermark can be treated as a kind of perturbation to manufacture Adversarial Examples (AEs) to fool Deep Neural Network (DNN) models. AEs refer to the adversaries deliberately crafting special inputs with perturbation to achieve malicious purposes [15], such as misclassification. For example, picture *a*) in Figure 1 can be identified as the Ground-Truth (GT) label “Vulture” by the trained InceptionV3. While after embedding the ACM logo in the top of host image, i.e., picture *d*), it is identified as “Kite” by the classifier. Given that the watermark logo itself is an independent semantic part rather than random noise, existing detection methods tend to be ineffective for watermark AE.

In this work, we propose Themis (“Themis” is a Greek deity of justice, divine order, and law), a novel watermark AE detection method via the *usable information theory* [2, 12]. It (also called ‘V-usable information’) is the variational extension of Shannon’s information theory [9], to take into account the modeling power

and computational constraints of the observer. Specifically, Themis neither modifies the protected classifier nor requires knowledge of the process for generating adversarial examples. It only needs to maintain an additional model (named *auxiliary model*) with the same structure as the protected classifier, but the parameter fitting is based on the empty input (e.g., all pixels are zeros). Essentially, the auxiliary model produces a probability distribution of labels. When an instance arrives, the output of the protected model is regarded as the *conditional entropy*, and the output of the auxiliary model is the *predictive entropy*. The subtraction of the predictive entropy and the conditional entropy is the amount of usable information, which will be used to judge whether this instance is an AE.

2 RELATED WORK

In this section, we investigate adversarial attacks, defense technology, and AE detection.

Adversarial Attack. The prior work of adversarial attack can be divided into white-box attacks [1] and black-box attacks. For the black-box attacks, common methods involve transfer-based attacks, score-based attacks, and decision-based attacks [15]. Furthermore, Adv-watermark [4] is an emerging black-box attack to generate watermark AE. Different from the traditional noise perturbation, Adv-watermark uses the logo watermark perturbation, bringing new challenges to AE detection, given logo watermark itself has independent semantics rather than noise. It leverages Basin Hopping Evolution to optimize settings for <watermark transparency, X coordinate, Y coordinate> during attack procedures¹.

Defense Technology. To resist adversarial attacks, some representative defensive techniques mainly focus on robust training, input transformation, randomization, and model ensemble [7, 14].

AE Detection. AE detection focuses more on performing analysis to find samples that may be AEs, so as to alert or refuse classification. These lines of related study can be categorized as *image transformation* [10], *reshaping feature* [11, 13], *data manifold analysis* [6], *introducing the rejection option* [8], *out-of-distribution detection* [5]. Five representative methods are used as baselines.

3 METHODOLOGY

Auxiliary Model Construction. Themis needs to construct an auxiliary model to obtain the predictive entropy. The auxiliary model essentially produces the probability distribution of the labels, so it only requires the GT label set $\mathcal{Y}$ from the training dataset. The auxiliary model f^* has the same network architecture as the protected model f , the difference is that the input is empty (all-zero pixels) when training f^* , i.e., the parameter of f^* is fitted by $(\emptyset, \mathcal{Y})$.

Usable Information Calculation. During the calculation, the auxiliary model f^* and the protected model f are responsible for obtaining the *predictive entropy* $H_{\mathcal{V}}(Y)$ and the *conditional entropy* $H_{\mathcal{V}}(Y|X)$. Subsequently, **V-information** refers to

$$I_{\mathcal{V}}(X \rightarrow Y) = H_{\mathcal{V}}(Y) - H_{\mathcal{V}}(Y|X) \quad (1)$$

The **pointwise V-information** (PVI) of an instance (x, y) is

$$PVI(x \rightarrow y) = -\log_2 f^*(y) + \log_2 f(y) \quad (2)$$

¹Particularly, Adv-watermark [4] is essentially an untargeted attack because it embeds the given watermark logo instead of the target image semantics. It is not yet known how to generate arbitrarily targeted AEs using specific watermarked logos, so our evaluation is based on untargeted attacks.

Detection Pipeline. For the instance set $\mathcal{S}$ to be tested, it could include samples of successful attacks and unsuccessful attacks, as well as clean samples. Our detection goal is to label AE or non-AE for each sample in $\mathcal{S}$. According to the outputs of f and f^* , each test sample will be calculated to the PVI value, and then compared with the preset threshold (based on the drop rate parameter d). If the PVI of the tested sample is less than the threshold, it should be tagged as “AE”, given it only has a small amount of usable information so the classifier is not confident enough. In contrast, those samples with PVI greater than the threshold are considered non-AEs.

4 EXPERIMENTS

We comprehensively evaluate Themis² in terms of the AE detection effect in various settings, time overhead, visualization, and elasticity for parameter choice.

Dataset and Setting. We randomly select 1,000 images from ImageNet [3] to conduct the related experiments. Meanwhile, we collect 5 conference logos (i.e., ACM, IEEE, USENIX, AAAI, Springer) for watermark perturbations. For the protected models, we choose two representative classification models with different structures: VGG19 and InceptionV3. About using watermark perturbation to generate AE, we keep the consistent parameter settings as in previous work (e.g., $scale = 1/4$, $iter = 3$, $NP = 50$, details reference Adv-watermark [4]). Particularly, the drop rate parameter d is set as 0.05 for all baselines and ours.

Table 1: Comparison experiments with 5 SOTA methods.

Inception	Accuracy ▲	Precision ▲	Recall ▲	FPR ▼
FS	45.97±4.09	87.08±3.80	12.44±1.18	2.73±0.42
MagNet	41.38±3.95	60.04±7.05	9.17±0.53	9.39±1.40
IT	78.65±0.88	81.31±2.77	83.73±1.18	29.30±1.43
RR	73.29±2.89	99.57±0.26	56.08±2.97	0.38±0.25
EBD	55.32±3.47	94.33±1.78	27.73±2.04	2.57±0.82
Themis	99.91±0.07	99.86±0.11	100.00±0.00	0.25±0.20
VGG19	Accuracy ▲	Precision ▲	Recall ▲	FPR ▼
FS	54.78±1.99	33.89±7.97	2.53±0.83	3.83±0.56
MagNet	53.99±1.97	39.60±1.97	7.66±0.34	9.25±0.01
IT	59.15±1.11	52.29±2.11	87.33±1.35	63.15±0.86
RR	76.71±0.99	98.44±0.75	48.07±1.50	0.64±0.38
EBD	65.80±1.45	87.07±2.02	26.55±1.99	3.16±0.68
Themis	96.12±0.56	98.94±0.36	92.18±1.29	0.59±0.30

Detection Effect. For 5 different perturbation watermarks, we report the mean results and standard deviation in Table 1. It is clear that Themis significantly outperforms the other five SOTA methods. Among them, FS and MagNet can rarely detect watermark AE (usually less than 10%), given the watermark perturbation is indeed different from the traditional noise perturbation. RR and EBD essentially utilize confidence or a variant of the softmax output, the results show that they lose a lot of precision for watermark AE detection. IT exhibits the best detection rate between these baselines, while an important problem is that the false positive rate of IT is too high, e.g., over 60% for VGG19 against conference logos. Overall, Themis achieves good performance with high recall/accuracy/precision and low false positive rate (FPR).

²<https://github.com/Secbrain/Themis>

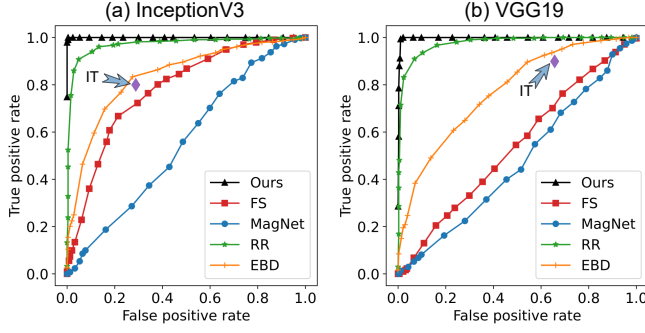


Figure 2: By setting different drop rate parameters d , we plot the ROC curve against the ACM logo perturbation.

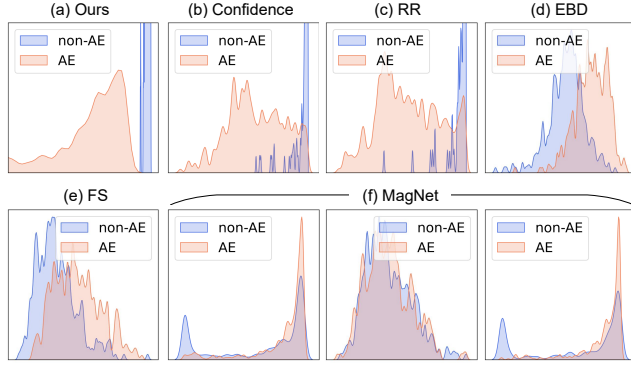


Figure 3: Using the kernel density estimate to visualize the distribution of observation metrics by various methods.

ROC Curve. We plot the detection’s ROC curve for Themis and baselines. Particularly, the detection effect of IT is not related to the parameter d , so it is expressed as a point. In Figure 2, the results correspond to using the ACM logo watermark to attack two victim classifiers. Regardless of which victim model, Themis achieves the largest Area Under Curve (AUC), indicating that Themis guarantees high TPR while only introducing low FPR.

Visualization. To provide more deep insights for Themis and existing AE detection methods, we leverage the kernel density estimate to plot the distribution of detection metrics between watermark AEs and non-AEs, based on the InceptionV3 classifier, against the ACM logo, *i.e.*, Figure 2 (a). Particularly, IT is not included because it employs image transformations to detect contradictory results. From Figure 3, we find that the usable information calculated by Themis is more distinguishable between watermark AEs and non-AEs (echoes back the design section), manifested in there is almost no intersection between the two parts and exists a clear dividing line in the middle. However, other methods’ metrics have some obviously visible intersections between the distribution space of AE and non-AE.

Overhead Analysis. We measure the time overhead for AE detection used by Themis and five baselines, and the results are summarized in Figure 4. All models run on the Ubuntu 18.04.2 server with Intel i7-12700K CPU, the NVIDIA TITAN Xp GPU, and 64 GB memory. Overall, Themis, IT, RR, and EBD are on one level (0.02s~0.04s), while FS and MagNet are on the other level (>0.1s). In addition, the auxiliary model and protected model in Themis

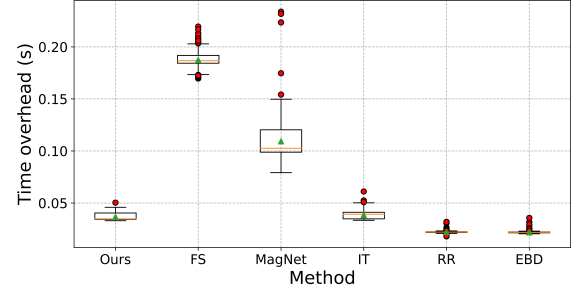


Figure 4: The time overhead of AE detection methods.

are independent of each other, so they can be parallelly placed on different devices in practice.

5 CONCLUSION

In this work, we propose Themis, a novel watermark AE detection scheme via the usable information theory. Themis neither modifies the protected classifier nor requires knowledge of the AE generation process. The empirical evaluations demonstrate that Themis can efficiently detect watermark AEs, and realize over 15% accuracy greater than five SOTA methods. Moreover, a series of additional experiments show that Themis has good elasticity for threshold settings and distinguishability between AEs & non-AEs.

ACKNOWLEDGMENTS

This work was supported in part by the National Natural Science Foundation of China (62227805, 62072398) and the National Key R&D Program of China (2020AAA0107700).

REFERENCES

- [1] Nicholas Carlini et al. Towards evaluating the robustness of neural networks. In *IEEE Symposium on Security and Privacy*, pages 39–57. IEEE Computer Society, 2017.
- [2] Kavin Ethayarajh et al. Understanding dataset difficulty with V-usable information. In *ICML*, volume 162 of *Proceedings of Machine Learning Research*, pages 5988–6008. PMLR, 2022.
- [3] ImageNet. Imagenet large scale visual recognition challenge 2012 (ilsrv2012). [EB/OL], 2021. <http://image-net.org/challenges/LSVRC/2012/2012-downloads> Accessed April 1, 2021.
- [4] Xiaojun Jia et al. Adv-watermark: A novel watermark perturbation for adversarial examples. In *ACM Multimedia*, pages 1579–1587. ACM, 2020.
- [5] Weitang Liu et al. Energy-based out-of-distribution detection. In *NeurIPS*, 2020.
- [6] Dongyu Meng and Hao Chen. Magnet: A two-pronged defense against adversarial examples. In *CCS*, pages 135–147. ACM, 2017.
- [7] Tianyu Pang et al. Improving adversarial robustness via promoting ensemble diversity. In *ICML*, volume 97 of *Proceedings of Machine Learning Research*, pages 4970–4979. PMLR, 2019.
- [8] Tianyu Pang et al. Two coupled rejection metrics can tell adversarial examples apart. In *CVPR*, pages 15202–15212. IEEE, 2022.
- [9] Claude E. Shannon. A mathematical theory of communication. *Bell Syst. Tech. J.*, 27(3):379–423, 1948.
- [10] Shixin Tian et al. Detecting adversarial examples through image transformation. In *AAAI*, pages 4139–4146. AAAI Press, 2018.
- [11] Weilin Xu et al. Feature squeezing: Detecting adversarial examples in deep neural networks. In *NDSS*. The Internet Society, 2018.
- [12] Yilun Xu et al. A theory of usable information under computational constraints. In *ICLR*. OpenReview.net, 2020.
- [13] Yijun Yang et al. What you see is not what the network infers: Detecting adversarial examples based on semantic contradiction. In *NDSS*. The Internet Society, 2022.
- [14] Hongyang Zhang et al. Theoretically principled trade-off between robustness and accuracy. In *ICML*, volume 97 of *Proceedings of Machine Learning Research*, pages 7472–7482. PMLR, 2019.
- [15] Ziming Zhao et al. SAGE: steering the adversarial generation of examples with accelerations. *IEEE Trans. Inf. Forensics Secur.*, 18:789–803, 2023.